

Problem Chosen

C

2026
MCM/ICM
Summary Sheet

Team Control Number

2632449

龚王博是我爹

Summary

Keywords:

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1 Introduction

1.1 Problem Background

1.2 Restatement of the Problem

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2

2.1

1. 11
2. 22
3. 33

2.2 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
E	entropy

*There are some variables that are not listed here and will be discussed in detail in each section.

3

3.1

3.1.1

3.1.2

3.2 Differential Equation Model Based on System Dynamics

The overall model can be expressed as:

3.3

3.3.1 Shannon Diversity Index

3.3.2

3.4

Table 2: Specified or Initial Values

Parameters or Variables	Values	Parameters or Variables	Values
r_P	0.1	β_A	0.1
k_G	1000	μ_A	0.05
α	0.05	β_B	0.1
r_W	0.5	μ_B	0.05
γ	0.05	P_0	100
η_I	0.0001	W_0	50
μ	0.05	I_0	30
μ'	0.02	A_0	1
η	0.05	B_0	1
μ_r	0.02	P_{r0}	20

Using Matlab, we calculated the temporal changes of these system dynamics variables in Figure 6.

And we calculated the eigenvalue of Jacobian Matrix:

$$\lambda(J) = -0.0911, -0.1111, 0.2300, 0.0400$$

4

4.1

4.2

1. **Enhancing Comprehensive Pest Control and Reducing Pesticide Dependence**
2. **Enhancing Ecosystem Resilience**
3. **Preventing Biodiversity Imbalance**

Table 3: Specified or Initial Values

Parameters or Variables	Values	Parameters or Variables	Values
η_I	0.005	η_{I_t}	0.005
α_P	1	α_W	15
γ	0.1	μ_r	1
μ_t	1	η_P	0.05
μ_{B_t}	0.005		

*The values of the variables not listed are the same as those in Table 2.

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5.1 Model Extension

5.2

5.2.1

5.2.2

5.2.3

- **Benefits:**
 - 1 **Ecological Benefits:**
 - 2 **Economic Benefits:**
 - 3 **Sustainability Benefits:**
- **Drawbacks**
 - 1 **Management Complexity:**
 - 2 **Ecological Risks:**

5.2.4

5.3

6 Strengths and Weaknesses

6.1 Strengths

- **Logic:**
- **Creativity:**
- **Credibility:**
- **Stability:**

6.2 Weaknesses

-
-

References

- [1] information on <https://en.tut tiempo.net/climate/cameroon.html>.

Letter

Report on Use of AI

1. OpenAI *ChatGPT*(<https://chat.openai.com/>)

Query1: Output:

Query2: Output:

Query3: Output: